

PE-EC603C — CMOS VLSI Design

Semester VI • 3L:0T:0P • 3 Credits

Module 4 Exam Notes: MOS & CMOS Inverters (Static Characteristics)

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1. CMOS Inverter Working & Static Behavior

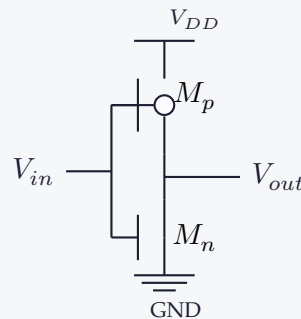
The static CMOS inverter is the core combinational building block of digital VLSI circuits.

1.1 Circuit Diagram and Operation

It consists of a pull-up pMOS transistor (connected between V_{DD} and V_{out}) and a pull-down nMOS transistor (connected between V_{out} and GND). **Operation:**

- $V_{in} = 0\text{ V}$ (Logic '0'): nMOS is cut-off; pMOS is ON (linear/saturated), pulling V_{out} up to V_{DD} (Logic '1').
- $V_{in} = V_{DD}$ (Logic '1'): pMOS is cut-off; nMOS is ON, pulling V_{out} down to 0 V (Logic '0').

CMOS Inverter Circuit & Equivalent Switch Representation



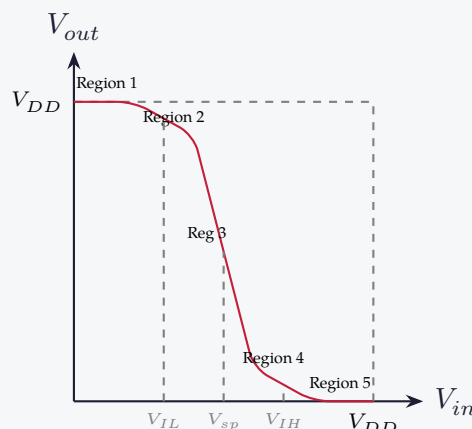
2. Voltage Transfer Characteristics (VTC)

The VTC plot of V_{out} vs. V_{in} characterizes the inverter's static response.

2.1 Five Regions of Operation

The VTC curve is divided into 5 distinct regions based on the state of the two transistors:

CMOS Inverter Voltage Transfer Characteristics (VTC)



1. **Region 1** ($0 \leq V_{in} < V_{thn}$): Input is low. nMOS is cut-off ($I_{Dn} = 0$). pMOS is ON in the linear region. $V_{out} = V_{DD}$.
2. **Region 2** ($V_{thn} \leq V_{in} < V_{sp}$): nMOS turns ON in saturation ($V_{DSn} > V_{GSn} - V_{thn}$). pMOS is ON in linear region. V_{out} begins to drop.
3. **Region 3** ($V_{in} = V_{sp}$): Switching threshold. Both nMOS and pMOS are in saturation. The output drops steeply.
4. **Region 4** ($V_{sp} < V_{in} \leq V_{DD} - |V_{thp}|$): nMOS is ON in linear region. pMOS is ON in saturation region.
5. **Region 5** ($V_{in} > V_{DD} - |V_{thp}|$): Input is high. pMOS is cut-off ($I_{Dp} = 0$). nMOS is ON in linear region. $V_{out} = 0$ V.

2.2 Derivation of Switching Threshold (V_{sp})

The switching threshold (or gate threshold voltage V_{sp}) is defined as the point where $V_{in} = V_{out}$. In Region 3, both transistors are in saturation:

$$I_{Dn} = I_{Dp} \Rightarrow \frac{1}{2}k_n (V_{sp} - V_{thn})^2 = \frac{1}{2}k_p (V_{DD} - V_{sp} - |V_{thp}|)^2 \quad (1)$$

Taking the square root on both sides:

$$\sqrt{k_n} (V_{sp} - V_{thn}) = \sqrt{k_p} (V_{DD} - V_{sp} - |V_{thp}|) \quad (2)$$

Divide by $\sqrt{k_n}$ and define the ratio $r = \sqrt{k_p/k_n}$:

$$V_{sp} - V_{thn} = r (V_{DD} - V_{sp} - |V_{thp}|) \quad (3)$$

$$V_{sp}(1 + r) = V_{thn} + r (V_{DD} - |V_{thp}|) \quad (4)$$

$$V_{sp} = \frac{V_{thn} + r (V_{DD} - |V_{thp}|)}{1 + r} \quad (5)$$

For a **symmetric inverter** where $k_n = k_p \Rightarrow r = 1$, and $V_{thn} = |V_{thp}|$:

$$V_{sp} = \frac{V_{DD}}{2} \quad (6)$$

To achieve a symmetric inverter ($k_n = k_p$), since electron mobility $\mu_n \approx 2.5\mu_p$, we must scale the channel widths:

$$\mu_n C_{ox} \frac{W_n}{L_n} = \mu_p C_{ox} \frac{W_p}{L_p} \Rightarrow \left(\frac{W}{L}\right)_p \approx 2.5 \left(\frac{W}{L}\right)_n \quad (7)$$

2.3 Noise Margins (NM_H, NM_L)

Noise margin measures the allowable electrical noise voltage on input terminals before the gate output state becomes corrupted.

Critical Voltages:

- V_{IL} (Input Low Voltage): The highest input voltage interpreted as Logic '0' (where $dV_{out}/dV_{in} = -1$).

- V_{IH} (Input High Voltage): The lowest input voltage interpreted as Logic '1' (where $dV_{out}/dV_{in} = -1$).
- V_{OL} (Output Low Voltage): Static low output: $V_{OL} = 0$ V for CMOS.
- V_{OH} (Output High Voltage): Static high output: $V_{OH} = V_{DD}$ for CMOS.

Formulas:

$$NM_L = V_{IL} - V_{OL} = V_{IL} \quad (8)$$

$$NM_H = V_{OH} - V_{IH} = V_{DD} - V_{IH} \quad (9)$$

3. Alternative load Inverters

Before static CMOS was standardized, alternative single-channel (nMOS-only) designs were used to avoid fabricating pMOS devices.

- **Resistive-Load Inverter:** Uses a passive resistor R_L as the pull-up load. Drawbacks: consumes a massive silicon area for high resistance; has static power dissipation when output is low ($I_{static} = V_{DD}/R_L$).
- **Enhancement-Load Inverter:** Uses an enhancement-mode nMOS transistor as a load (gate connected to V_{DD} or drain). Drawbacks: Output high voltage is degraded ($V_{OH} = V_{DD} - V_{thn}$), and suffers from static power dissipation when V_{out} is low.
- **Depletion-Load Inverter:** Uses a depletion-mode nMOS transistor (with gate shorted to source, $V_{GS} = 0$) as a load. Acts as a constant current source, yielding a sharper VTC transition and full logic swing ($V_{OH} = V_{DD}$), but still has high static power dissipation when output is low.

CMOS vs Depletion-Load Inverter Comparison

Feature	Static CMOS Inverter	Depletion-Load Inverter
Device Count	2 (1 nMOS, 1 pMOS).	2 (1 enhancement, 1 depletion nMOS).
Logic Swing	Full rail-to-rail swing (0 to V_{DD}).	High swing (nearly 0 to V_{DD}).
Static Power	Zero (ignoring tiny leakage).	High (when V_{out} is low).
Transition Width	Narrow, sharp drop.	Moderate.
Fabrication	High complexity (well formation).	Low complexity (single-channel).

Quick Revision — Module 4

MOS & CMOS Inverters (Static Characteristics) | PE-EC603C

Key Inverter Equations

$$V_{sp} = \frac{V_{thn} + r(V_{DD} - |V_{thp}|)}{1 + r} \quad \text{where } r = \sqrt{\frac{k_p}{k_n}} = \sqrt{\frac{\mu_p W_p L_n}{\mu_n W_n L_p}}$$

$$\text{Symmetric Inverter Width Ratio: } \left(\frac{W}{L}\right)_p \approx 2.5 \left(\frac{W}{L}\right)_n \quad V_{sp} = \frac{V_{DD}}{2}$$

$$NM_L = V_{IL} - V_{OL} = V_{IL} \quad NM_H = V_{OH} - V_{IH} = V_{DD} - V_{IH}$$

One-Line Definitions

- **VTC:** Plot of output voltage against input voltage characterizing static gain and noise immunities.
- **Switching Threshold (V_{sp}):** Input voltage at which the output voltage equals the input voltage ($V_{in} = V_{out}$).
- **Noise Margin:** The maximum noise voltage that can be added to the input signal without corrupting the output logic state.
- **Symmetric Inverter:** An inverter with identical pull-up and pull-down strengths, placing V_{sp} exactly at $V_{DD}/2$.
- **Static Power Dissipation:** Power consumed by the gate when inputs are static (non-switching).
- **Depletion Load:** An nMOS transistor with $V_{th} < 0$, configured with gate shorted to source to act as a current source.

Frequently Asked / Viva Questions

Q1. Why is the static power dissipation of a CMOS inverter practically zero?

Because in either steady state (Input = '0' or Input = '1'), one of the two series transistors (either pMOS or nMOS) is completely cut off, presenting an open circuit. Consequently, no direct path exists from V_{DD} to GND, and current flow is limited to subthreshold and junction leakages (picoamps).

Q2. Derive the switching threshold V_{sp} of a CMOS inverter.

Equating saturation currents $I_{Dn} = I_{Dp} \Rightarrow \frac{1}{2}k_n(V_{sp} - V_{thn})^2 = \frac{1}{2}k_p(V_{DD} - V_{sp} - |V_{thp}|)^2$. Taking the square root, collecting terms, and solving for V_{sp} yields $V_{sp} = \frac{V_{thn} + r(V_{DD} - |V_{thp}|)}{1 + r}$ where $r = \sqrt{k_p/k_n}$.

Q3. How do we design a symmetric CMOS inverter and why is it preferred?

A symmetric inverter has $V_{sp} = V_{DD}/2$, which maximizes both noise margins ($NM_L = NM_H$) and balances rise and fall propagation delays. It is achieved by

Frequently Asked / Viva Questions (contd.)

setting $k_n = k_p \Rightarrow \mu_n(W/L)_n = \mu_p(W/L)_p \Rightarrow (W/L)_p \approx 2.5(W/L)_n$.

Q4. Define Noise Margin and write its formulas for a CMOS inverter.

Noise margin represents the gate's noise immunity. For CMOS, since $V_{OL} = 0$ and $V_{OH} = V_{DD}$, the margins are $NM_L = V_{IL} - V_{OL} = V_{IL}$ and $NM_H = V_{OH} - V_{IH} = V_{DD} - V_{IH}$, where V_{IL} and V_{IH} are input voltages where the VTC slope $dV_{out}/dV_{in} = -1$.

Q5. Explain the VTC region 3 of a CMOS inverter.

In Region 3, the input is close to $V_{DD}/2$. Both nMOS and pMOS are in saturation simultaneously, acting as high-impedance current sources. This creates a very high voltage gain ($dV_{out}/dV_{in} \ll -1$), causing a steep drop in V_{out} .

Q6. What are the disadvantages of a resistive-load inverter?

First, it consumes static power ($P = V_{DD}^2/R_L$) whenever the output is low because the pull-down nMOS is ON. Second, to get low V_{OL} , R_L must be very large (tens of $k\Omega$), which requires a massive surface area on silicon compared to a transistor.

Q7. Why does an enhancement-load inverter suffer from degraded V_{OH} ?

In an enhancement-load nMOS inverter, the load transistor turns OFF when the output rises to $V_{DD} - V_{thn}$ because its gate-source voltage drops below the threshold voltage. Thus, it cannot pull the output all the way up to V_{DD} .

Q8. Explain the depletion-load inverter and its advantage over resistive-load.

The depletion-load inverter uses a depletion-mode nMOS with $V_{th} < 0$ and gate shorted to source ($V_{GS} = 0$). It behaves as a constant current source rather than a passive resistor, which pulls the output up to the full V_{DD} and yields a sharper VTC transition.

Q9. If a CMOS inverter has $V_{DD} = 3.3\text{ V}$, $V_{thn} = 0.7\text{ V}$, $|V_{thp}| = 0.8\text{ V}$, and $r = 1.0$, calculate V_{sp} .

Using the formula: $V_{sp} = \frac{V_{thn} + r(V_{DD} - |V_{thp}|)}{1+r} = \frac{0.7 + 1.0 \times (3.3 - 0.8)}{1+1} = \frac{0.7 + 2.5}{2} = 1.6\text{ V}$.

Q10. How does noise margin scale when we reduce the supply voltage V_{DD} ?

As V_{DD} scales down, the total transition range of the VTC shrinks. Since critical points V_{IL} and V_{IH} scale proportionally with V_{DD} , the absolute noise margins (NM_L, NM_H) decrease, making the circuit more sensitive to external electrical noise.